

Effects of intermittent running session on eccentric hamstrings strength: case study

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Abstract: This study investigates the effects produced by intermittent running session on eccentric hamstrings strength. Data obtained from pre-test and post-test (separated by intermittent running session) were compared. Eccentric hamstring strength was evaluated by performing a Nordic Hamstring exercise. Results showed a 17.5% decrease of maximal hamstring force between pre-test and post-test. In conclusion, intermittent running session at a submaximal running speed induced a decrease in hamstring maximal force. Therefore, PhyNord (Phyling, France) could be used by physical trainers to assess the impact of a training session.

1. Introduction

Despite the development of training monitoring tools, hamstrings injuries are the most common injuries in activities involving high speed running [1]. The development of maximum hamstring strength is essential to reduce the risk of injury [2]. Eccentric hamstring force monitoring is usually performed by instrumented devices such as the PhyNord (Phyling, France). These tools provide information not only on maximal eccentric strength but also on limb imbalances and asymmetries [2]. Nordic hamstring exercise is widely used to assess hamstring strength and improve muscles force [3]. This exercise consists of kneeling at a 90° knee angle and lean forward to the horizontal position, holding on as long as possible. Ankles are locked by the force sensors. A trial is considered valid if a maximum of force is reached and followed by a rapid decrease in force. Fatigue can be assessed by the value of maximum force generated by hamstrings. Intermittent running at submaximal velocity induces peripheral and central fatigue [4] which leads to running kinematics and forces variations [5]. The aim of this study was to identify if maximum hamstring could be influenced by intermittent running.



Figure 1: Starting position of Nordic Hamstring exercise on the PhyNord

2. Method

2.1. Material and Participant

A PhyNord (Phyling, France) was used to measure the eccentric force developed by hamstrings. It is composed of an aluminum bench and force sensors holding ankles to perform the Nordic hamstring exercise (Figure 1). The data acquisition frequency was set to 200Hz. The data analysis was realized by a Maxi-Hub (Phyling, France), which gives access to real-time feedback and automatic analysis. The running session was recorded by a GPS watch Forunner 745 (Garmin, USA) to control the fatigue protocol. One participant (1.78m, 70kg) was recruited for this study. He was free from lower limb injury and runs over 20km per week.

2.2. Protocol

The fatigue protocol consisted in an intermittent running session beginning by 25 min warm-up at a self-selected pace. The participant was then asked to repeat five 500m running bouts at maximal aerobic velocity ($v\text{VO}_{2\text{max}}$) separated with 200m bouts at the warm-up velocity.

Maximum hamstring strength was assessed before (pre-test) and after (post-test) the intermittent running. For both tests, participant performed 3 Nordic Hamstring repetitions (after a five minutes muscle warm-up).

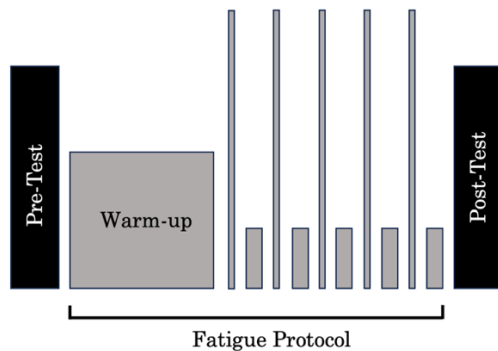


Figure 2: Diagram of experimental procedure

2.3. Data analysis

The maximum of right (RF) and left (LF) maximal hamstring eccentric force was calculated by the Maxi-Hub. Raw data are presented in Figure 3. Data from the best of the three repetitions was kept for pre-post comparison.

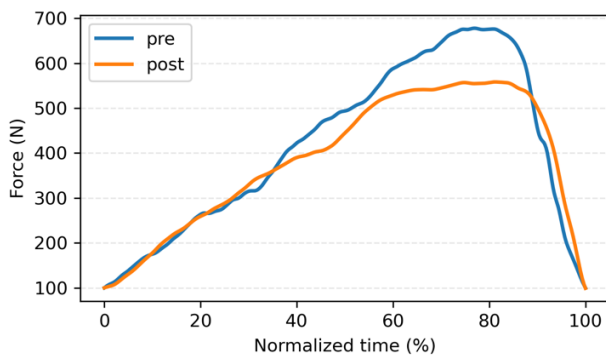


Figure 3: Raw data from the best repetition of pre-test and post-test.

3. Results and discussion

Test	Values	
	Pre	Post
Max Ftot	677	558
Max RF (N)	314	280
Max LF (N)	362	278
Imbalance Ratio (%)	7.1	0.4

Table 1: Maximum values of total force (F_{tot}), right force (RF) and left force (LF) and imbalance ratio for Pre and Post-test.

The maximum force decreased from 677N to 558N between pre-test and post-test. This corresponds to a decrease of performance of 17.5%. Results shows that intermittent running session at submaximal velocities leads to hamstrings fatigue. The addition of electromyograms could help to better identify which fatigue mechanism is involved. The PhyNord appears as a practical on-field tool to assess athlete's muscular fatigue.

4. Bibliography

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